

Gyroscopic Effect on Automobile

by

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What is Gyroscope?

Definition:

A gyroscope is a device for measuring or maintaining orientation, based on the principles of the conservation of angular momentum.

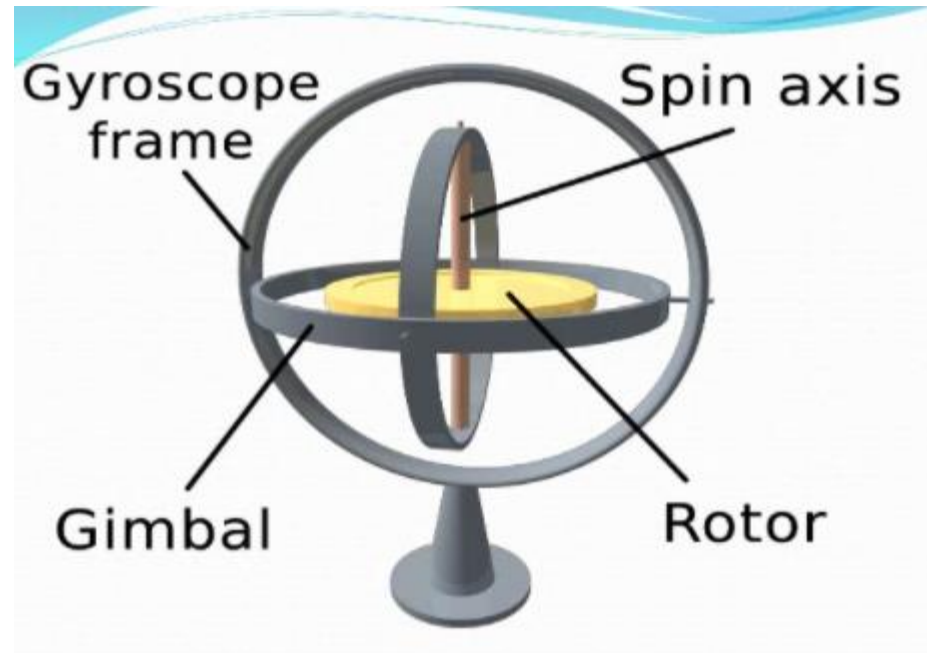


Fig. 1. Gyroscope

- If the axis of spinning or rotating body is given an angular motion about an axis perpendicular to the axis of spin, an angular acceleration acts on the body about the third perpendicular axis.
- The torque required to produce this acceleration is known as the active gyroscopic torque.
- A reactive gyroscopic torque or couple also acts similar to the concept of centripetal and centrifugal forces on a rotating body.
- The effect produced by the reactive gyroscopic couple is known as the gyroscopic effect.
- Thus aeroplane, ships, automobiles, etc, that have rotating parts in the form of wheels or rotors of engines experiences gyroscopic effect while taking turn, such as when axes of spin is subjected to some angular motions.

Gyroscope contd...

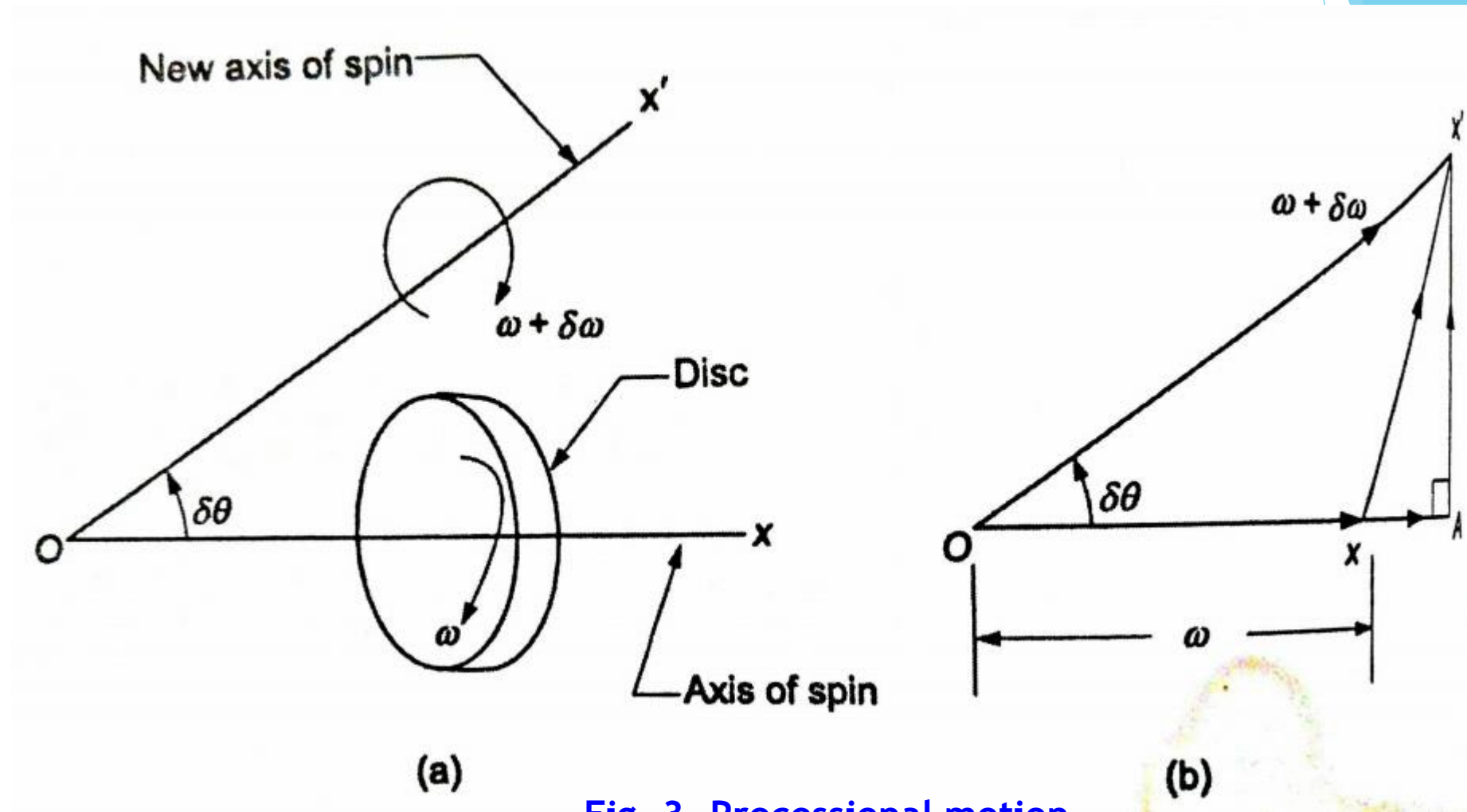


Fig. 3. Precessional motion

Gyroscope contd...

Angular acceleration along

$$\begin{aligned} OX &= \lim_{\delta t \rightarrow 0} \left[\frac{(\omega + \delta\omega) \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{\omega \cos \delta\theta + \delta\omega \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{\omega + \delta\omega - \omega}{\delta t} \right] \quad (\text{Because } \cos \delta\theta \approx 1) \\ &= \lim_{\delta t \rightarrow 0} \frac{\delta\omega}{\delta t} \\ &= \frac{d\omega}{dt} \end{aligned}$$

Gyroscope contd...

Angular (or gyroscopic) acceleration perpendicular to OX ,

$$\begin{aligned} &= \lim_{\delta t \rightarrow 0} \left[\frac{(\omega + \delta\omega) \sin \delta\theta}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \frac{(\omega \cdot \delta\theta + \delta\omega \cdot \delta\theta)}{\delta t} \\ &= \frac{\omega \cdot d\theta}{dt} \quad [\text{Neglecting } \delta\omega \cdot \delta\theta, \text{ being small}] \\ &= \omega \cdot \omega_p \end{aligned}$$

where ω_p is the precessional angular speed of the spin axis.
Total angular acceleration of disc = XX'



Gyroscope contd...

$$\begin{aligned} &= XA \quad AX' \\ &= \frac{d\omega}{dt} \quad \omega \cdot \frac{d\theta}{dt} \\ &= \frac{d\omega}{dt} \quad \omega \cdot \omega_p \end{aligned}$$

where $\rightarrow$ represents vector sum.

Gyroscope contd...

Total angular acceleration along XX' may be written as:

$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

1. $d\omega/dt$, represents change in the magnitude of the angular velocity of the rotor.
2. $\omega.d\theta/dt$, represents change in the direction of the axis of spin, the direction of AX' is from A to X' in the vector diagram (being a component of XX'), the acceleration acts clockwise in the vertical plane xz (when viewed from front along y -axis).

Gyroscope contd...

Fundamentals of Gyroscopic Motion:

Gyroscope: It is a body which spinning about an axis is free to move in other direction under the action of external forces.

Axis of Spin: It is axis about which the body revolves.

Gyroscopic effect: consider a body spinning about an axis OX . If couple represented by vector OZ perpendicular to OX .

- Then the body tries to precess about an axis OY , which is perpendicular to both OX and OZ .
- This combined effect is called gyroscopic or precessional effect.
- The plane of spin, plane of precession, and plane of gyroscopic couple are mutually perpendicular.

Gyroscope contd...

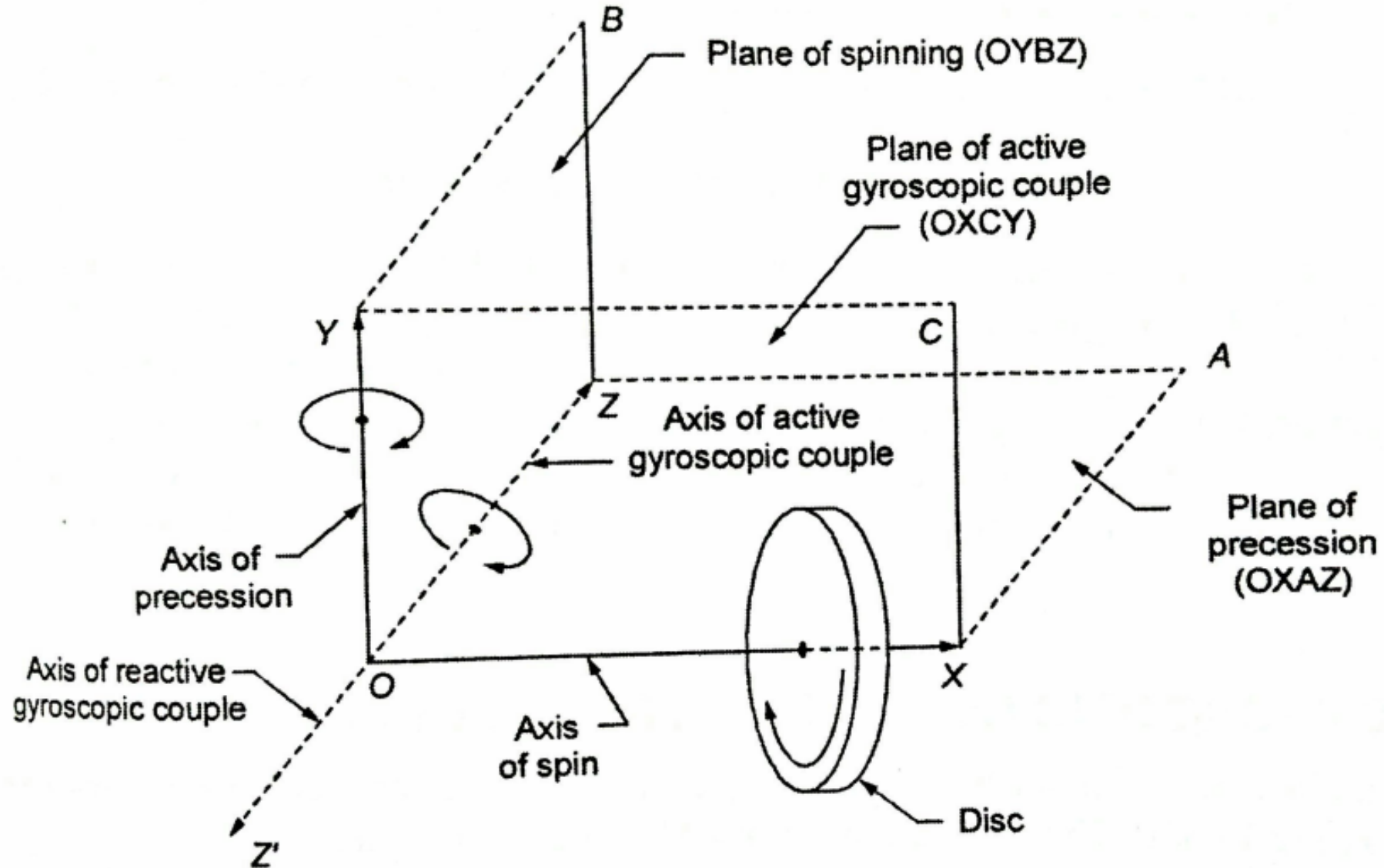


Fig. 4. Gyroscopic effect

Gyroscope contd...

Gyroscopic Couple:

Torque or Couple C:

$$T = I \frac{d\omega}{dt} + T\omega \frac{d\theta}{dt}$$

$$T = I\alpha + T\omega\omega_p$$

$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

Gyroscopic Couple

Gyroscopic Effect on Automobile Vehicle

Stability of Four Wheeled Vehicle negotiating a turn.



Stable condition

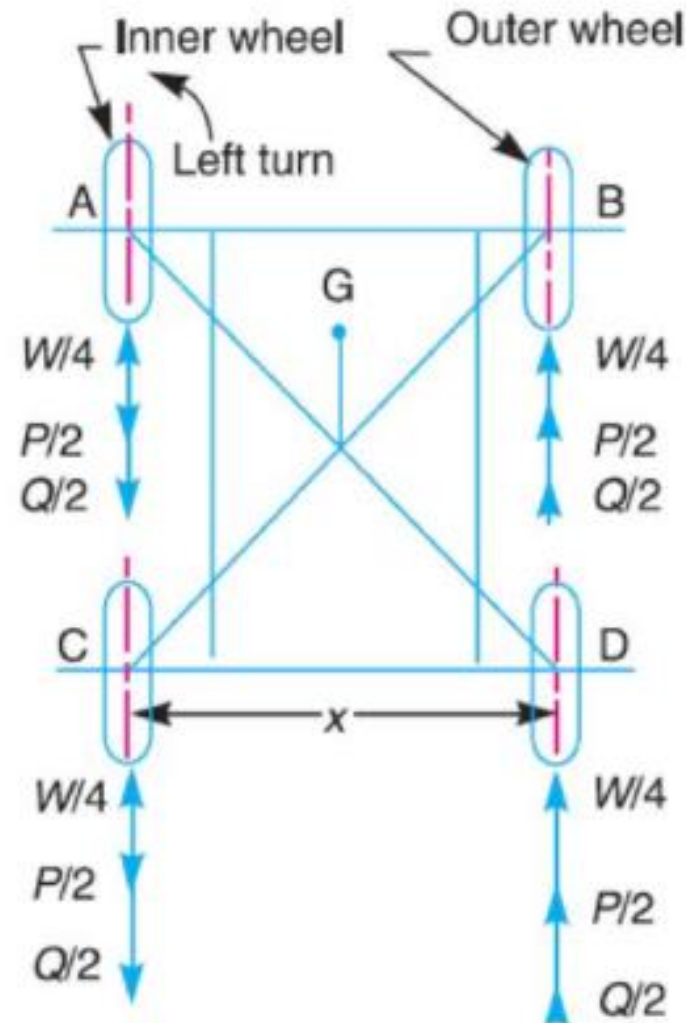


Unstable Condition

Stability of a Four Wheel Drive Moving in a Curved Path

- Consider the four wheels A , B , C and D of an automobile locomotive taking a turn towards left as shown in Fig.
- The wheels A and C are inner wheels, whereas B and D are outer wheels. The centre of gravity (C.G.) of the vehicle lies vertically above the road surface.

Let m = Mass of the vehicle in kg,
 W = Weight of the vehicle in newtons = $m.g$,
 r_w = Radius of the wheels in metres,
 R = Radius of curvature in metres
 $(R > r_w)$,
 h = Distance of centre of gravity, vertically above the road surface in metres,
 x = Width of track in metres,
 I_w = Mass moment of inertia of one of the wheels in kg-m^2 ,



ω_W = Angular velocity of the wheels or velocity of spin in rad/s,

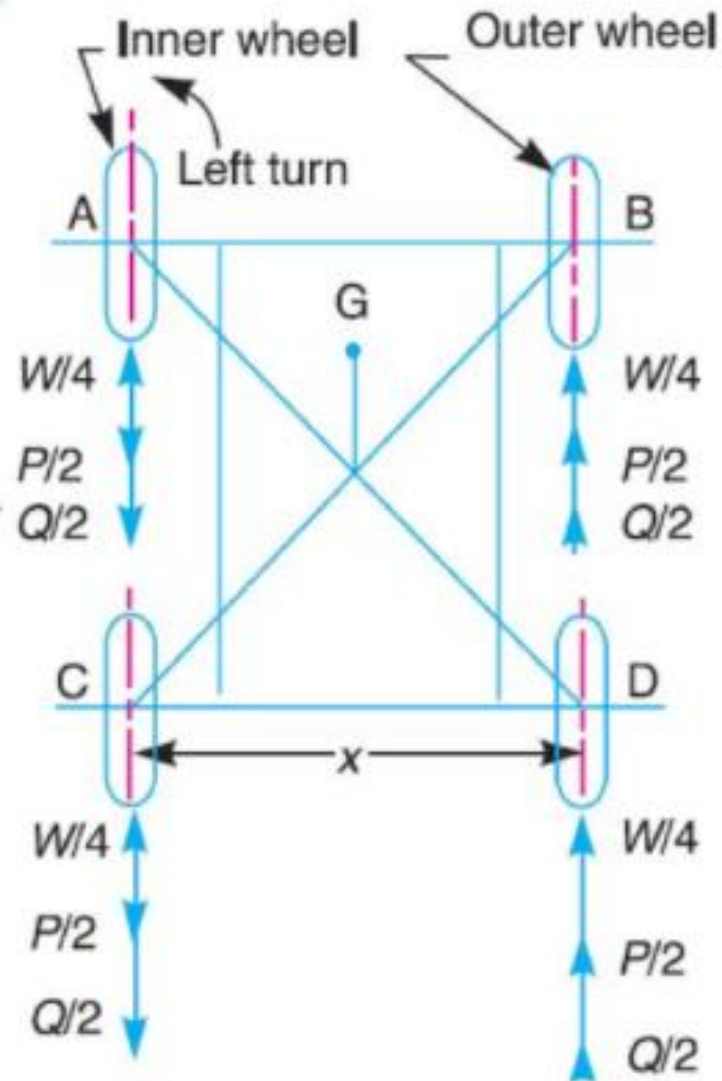
I_E = Mass moment of inertia of the rotating parts of the engine in kg-m^2 ,

ω_E = Angular velocity of the rotating parts of the engine in rad/s,

G = Gear ratio = ω_E / ω_W ,

v = Linear velocity of the vehicle in m/s = $\omega_W \cdot r_W$

- A little consideration will show, that the weight of the vehicle (W) will be equally distributed over the four wheels which will act downwards.
- The reaction between each wheel and the road surface of the same magnitude will act upwards.
- Therefore Road reaction over each wheel, = $W/4 = m \cdot g / 4$ newtons



- Let us now consider the effect of the gyroscopic couple and centrifugal couple on the vehicle.

1. Effect of the gyroscopic couple

Since the vehicle takes a turn towards left due to the precession and other rotating parts, therefore a gyroscopic couple will act.

We know that velocity of precession,

$$\omega_p = v/R$$

$\therefore$ Gyroscopic couple due to 4 wheels,

$$C_W = 4 I_W \cdot \omega_W \cdot \omega_p$$

and gyroscopic couple due to the rotating parts of the engine,

$$C_E = I_E \cdot \omega_E \cdot \omega_p = I_E \cdot G \cdot \omega_W \cdot \omega_p \quad \dots (\because G = \omega_E / \omega_W)$$

$\therefore$ Net gyroscopic couple,

$$\begin{aligned} C &= C_W \pm C_E = 4 I_W \cdot \omega_W \cdot \omega_p \pm I_E \cdot G \cdot \omega_W \cdot \omega_p \\ &= \omega_W \cdot \omega_p (4 I_W \pm G \cdot I_E) \end{aligned}$$

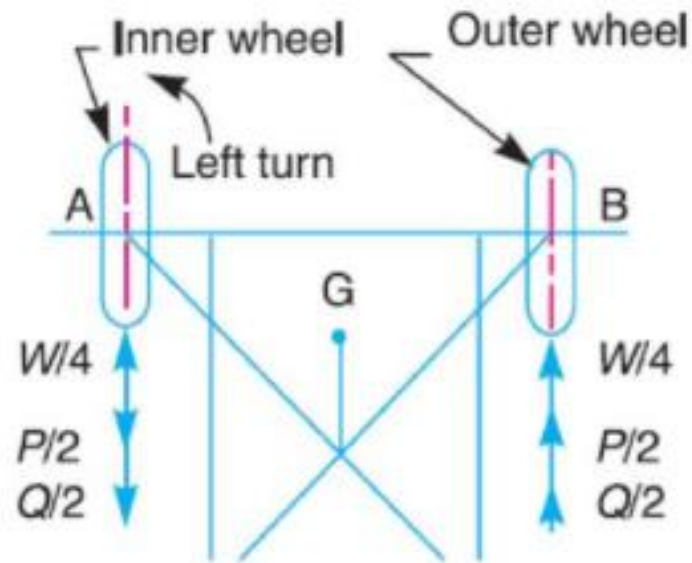
The **positive sign is used when the wheels and rotating parts of the engine rotate in the same** direction. If the rotating parts of the engine revolves in opposite direction, then **negative sign is used**.

- Due to the gyroscopic couple, vertical reaction on the road surface will be produced.
- The reaction will be vertically upwards on the outer wheels and vertically downwards on the inner wheels.
- Let the magnitude of this reaction at the two outer or inner wheels be P newtons. Then
- $P \times x = C$ or $P = C/x$
- Vertical reaction at each of the outer or inner wheels, $P/2 = C/2x$

Note: when rotating parts of the engine rotate in opposite directions, then -ve sign is used, i.e. net gyroscopic couple,

$$C = C_W - C_E$$

When $C_E > C_W$, then C will be -ve. Thus the reaction will be vertically downwards on the outer wheels and vertically upwards on the inner wheels.



2. Effect of the centrifugal couple

- Since the vehicle moves along a curved path, therefore centrifugal force will act outwardly at the centre of gravity of the vehicle.
- The effect of this centrifugal force is also to overturn the vehicle.
- We know that centrifugal force,

$$F_C = \frac{m \times v^2}{R}$$

∴ The couple tending to overturn the vehicle or overturning couple,

$$C_O = F_C \times h = \frac{m.v^2}{R} \times h$$

- This overturning couple is balanced by vertical reactions, which are vertically upwards on the outer wheels and vertically downwards on the inner wheels.

$$Q \times x = C_O \quad \text{or} \quad Q = \frac{C_O}{x} = \frac{m.v^2.h}{R.x}$$

∴ Vertical reaction at each of the outer or inner wheels,

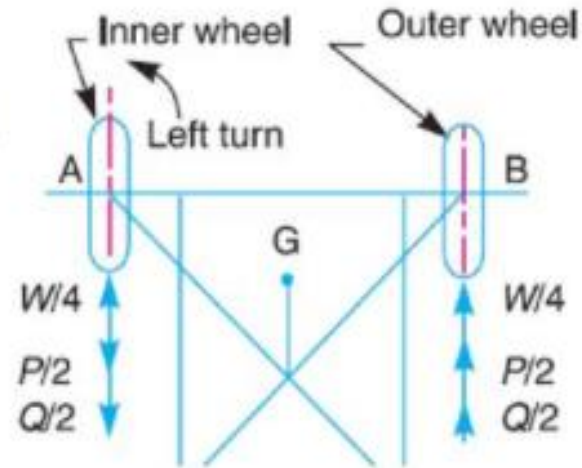
$$\frac{Q}{2} = \frac{m.v^2.h}{2R.x}$$

∴ Total vertical reaction at each of the outer wheel,

$$P_O = \frac{W}{4} + \frac{P}{2} + \frac{Q}{2}$$

total vertical reaction at each of the inner wheel,

$$P_I = \frac{W}{4} - \frac{P}{2} - \frac{Q}{2}$$



- A little consideration will show that when the vehicle is running at high speeds, P_I may be zero or even negative. This will cause the inner wheels to leave the ground thus tending to overturn the automobile. In order to have the contact between the inner wheels and the ground, the sum of $P/2$ and $Q/2$ must be less than $W/4$.

Problem 1. Each wheel of a four-wheeled rear engine automobile has a moment of inertia of 2.4 kg-m^2 and an effective diameter of 660 mm. The rotating parts of the engine have a moment of inertia of 1.2 kg-m^2 . The gear ratio of engine to the back wheel is 3 to 1. **The engine axis is parallel to the rear axle and the crankshaft rotates in the same sense as the road wheels.** The mass of the vehicle is 2200 kg and centre of the mass is 550 mm above the road level. The track width of the vehicle is 1.5 m. **Determine the limiting speed of the vehicle around a curve with 80 m radius so that all the four wheels maintain contact with the road surface.**

Solution: Moment of inertia of each wheel $I_w = 2.4 \text{ kg-m}^2$

Mass of the vehicle is $m = 2200 \text{ kg}$

An effective diameter of $d = 660 \text{ mm}$; thus radius $r = 0.330 \text{ m}$

Moment of inertia of engine $I_e = 1.2 \text{ kg-m}^2$

Centre of the mass is $h = 550 \text{ mm}$ above the road level

The track width of the vehicle is $w = 1.5$ m.

A curve with radius $R = 80$ m, $G = \omega_e / \omega_w = 3$

1. Reaction due to weight:

$$R_w = \frac{mg}{4} = \frac{2200 \times 9.81}{4} = 5395.5 \text{ N (upwards)}$$

2. Reaction due to gyroscopic couple:

gyroscopic couple due to wheel: $C_w = 4I_w \frac{v^2}{rR} = 4 \times 2.4 \times \frac{v^2}{0.33 \times 80} = 0.364v^2$

$$\omega_w = \frac{v}{r}; \omega_p = \frac{v}{R}$$

gyroscopic couple due to engine: $C_e = I_e G \omega_w \omega_p = 1.2 \times 3 \times \frac{v^2}{0.33 \times 80} = 0.136v^2$

$$C_G = C_w + C_e = 0.364v^2 + 0.136v^2 = 0.5v^2$$

Reaction on each outer wheel:

$$R_{Go} = \frac{C_G}{2w} = \frac{0.5v^2}{2 \times 1.5} = 0.167v^2 (\text{upwards})$$

Reaction on each inner wheel:

$$R_{Gi} = \frac{C_G}{2w} = \frac{0.5v^2}{2 \times 1.5} = 0.167v^2 (\text{downwards})$$

3. Reaction due to centrifugal couple:

$$C_c = \frac{mv^2}{R} h = \frac{2200 \times v^2}{80} \times 0.55 = 15.125v^2$$

Reaction on each outer wheel: $R_{co} = \frac{C_e}{2w} = \frac{15.125v^2}{2 \times 1.5} = 5.042v^2 (\text{upwards})$

Reaction on each inner wheel: $R_{ci} = 5.042v^2 (\text{downwards})$

For maximum safe speed, the condition is:

$$R_w = R_{Gi} + R_{ci}$$

$$5395.5 = (0.167 + 5.042)v^2$$

$$v^2 = 1035.8$$

$$v = 32.18m / sec$$

$$v = \frac{32.18 \times 3600}{1000} = 115.9km / h$$

Stability of Two Wheeler negotiating a turn

- Fig. shows a two wheeler vehicle taking **left turn** over a curved path. The vehicle is inclined to the vertical for equilibrium by an angle θ known as angle of heel.



Let

m = Mass of the vehicle and its rider in kg,

W = Weight of the vehicle and its rider in

newtons = $m.g$,

h = Height of the centre of gravity of the vehicle and rider,

r_W = Radius of the wheels,

R = Radius of track or curvature,

I_W = Mass moment of inertia of each wheel,

I_E = Mass moment of inertia of the rotating parts of the engine,

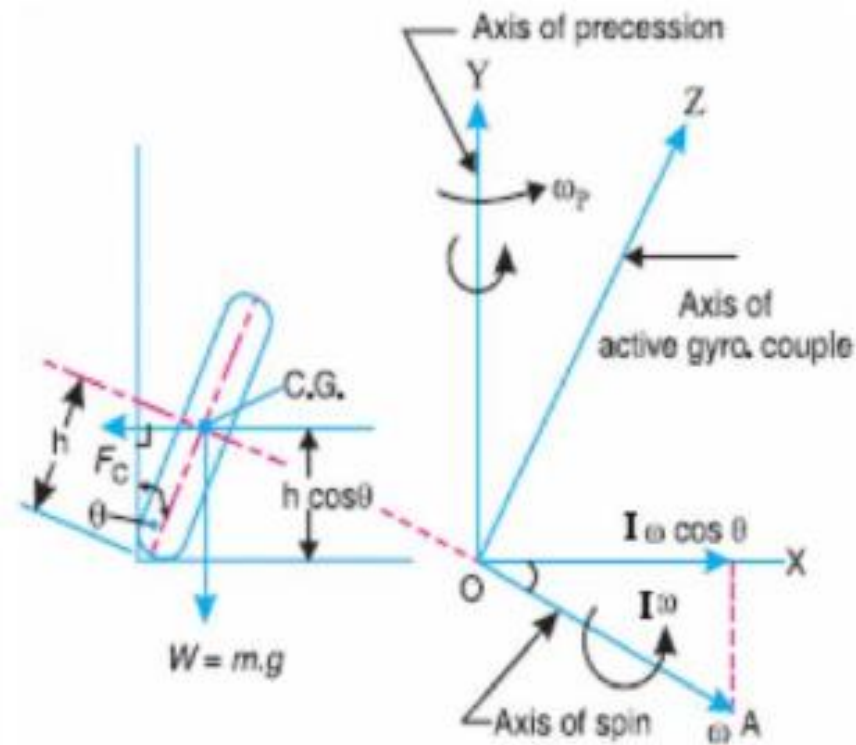
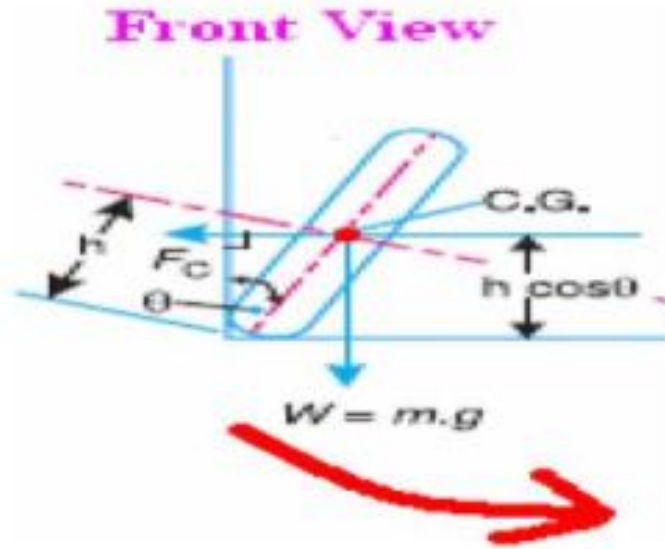
ω_W = Angular velocity of the wheels,

ω_E = Angular velocity of the engine rotating parts,

G = Gear ratio = ω_E / ω_W ,

v = Linear velocity of the vehicle = $\omega_W \times r_W$,

θ = Angle of heel. It is inclination of the vehicle to the vertical for equilibrium



1. Effect of Gyroscopic Couple

$$V = \omega_w \times r_w$$

$$\omega_E = G \cdot \omega_W \text{ or } \omega_E = G \cdot v / r_w$$

$$\text{Angular momentum due to wheels} = 2 I_w \omega_w$$

$$\text{Angular momentum due to engine and transmission} = I_E \omega_E$$

$$\begin{aligned} \text{Total angular momentum (} I \times \omega \text{)} &= 2 I_w \omega_w \pm I \omega_E \\ &= \frac{v}{r_w} (2 I_w \pm G I_E) \end{aligned}$$

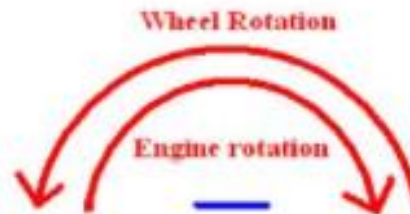
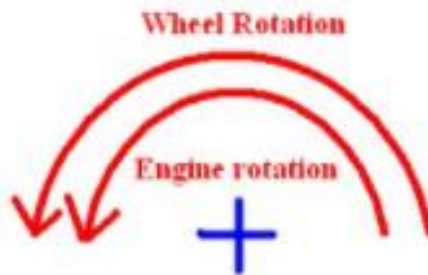
$$\text{Velocity of precession} = \omega_p = \frac{v}{R}$$

- It is observed that, when the wheels move over the curved path, the vehicle is always inclined at an angle θ with the vertical plane as shown in Fig. This angle is known as 'angle of heel'.
- Gyroscopic Couple,

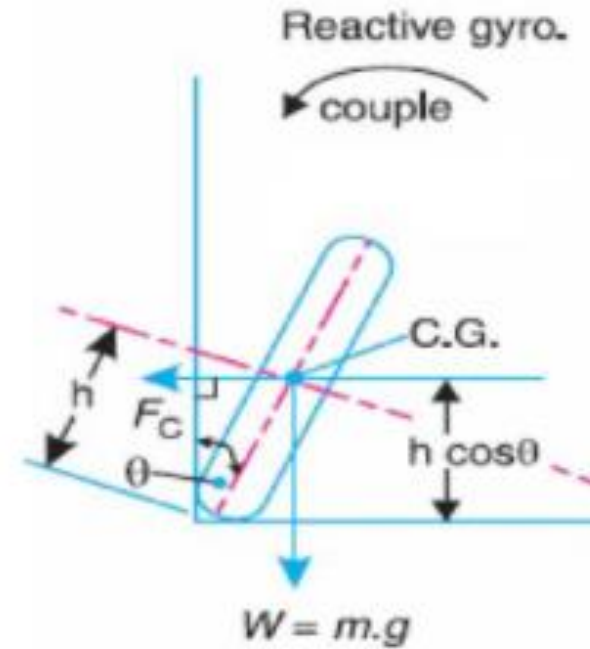
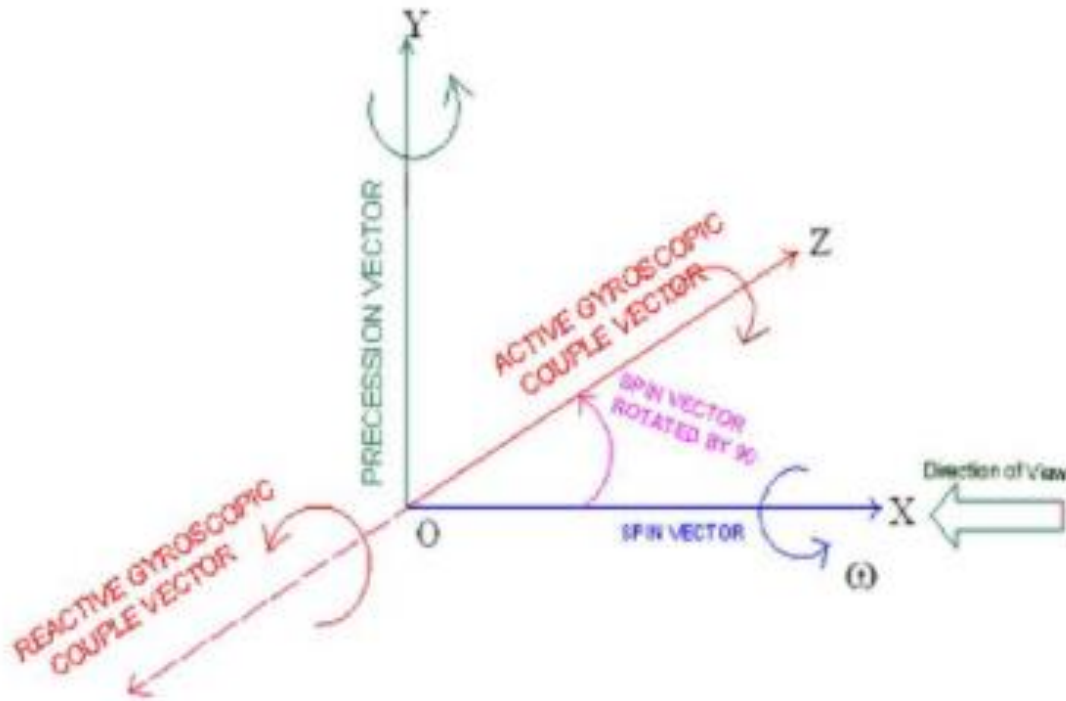
$$C_g = (I\omega) \cos\theta \times \omega_p$$

$$C_g = \frac{v^2}{Rr_w} (2I_w \pm GI_e) \cos\theta$$

Note: When the engine is rotating in the same direction as that of wheels, then the positive sign is used in the above equation. However, if the engine rotates in opposite direction to wheels, then negative sign is used.



- The gyroscopic couple will act over the vehicle outwards i.e., in the anticlockwise direction when seen from the front of the two wheeler. This couple tends to overturn/topple the vehicle in the outward direction as shown in Fig.



2. Effect of Centrifugal Couple

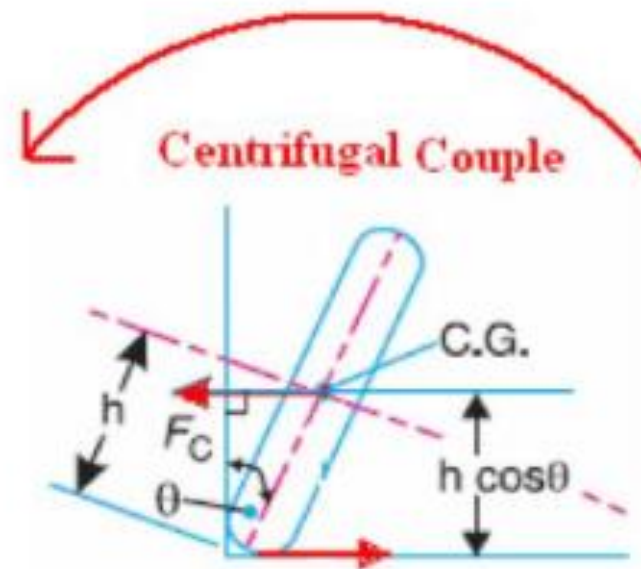
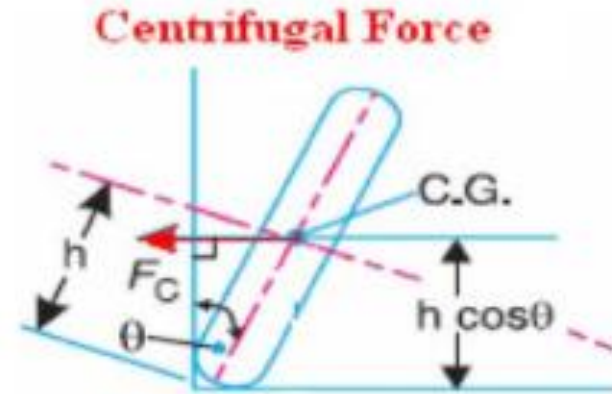
Centrifugal force,

$$F_c = \frac{mv^2}{R}$$

Centrifugal Couple,

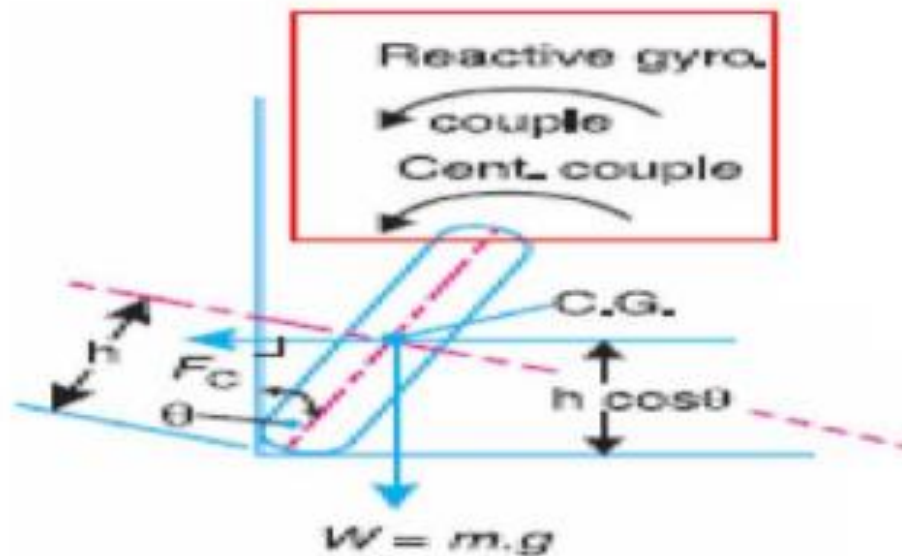
$$C_c = F_c \times h \cos \theta$$
$$= \frac{mv^2}{R} h \cos \theta$$

- The Centrifugal couple will act over the two wheeler outwards i.e., in the anticlockwise direction when seen from the front of the two wheeler. This couple tends to overturn/topple the vehicle in the outward direction



- total Over turning couple: $C = C_g + C_c$

$$C = \frac{v^2}{Rr} (2I_w + GI_e) \cos \theta + \frac{mv^2}{R} h \cos \theta$$

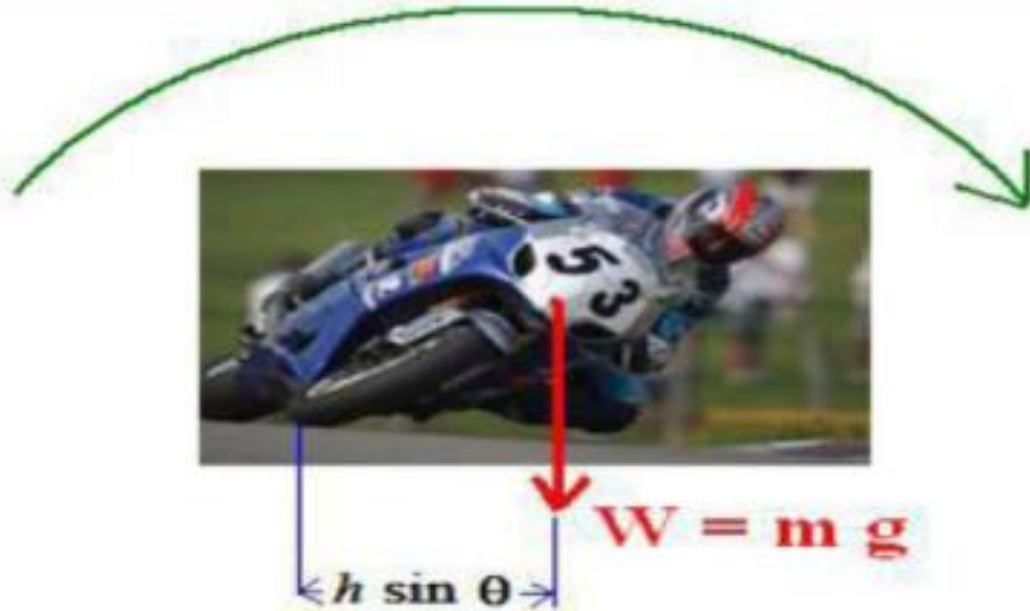


- For the vehicle to be in equilibrium, overturning couple should be equal to balancing couple acting in clockwise direction due to the weight of the vehicle and rider.

$$C = mgh \sin \theta$$

For the stability, overturning couple must be equal to balancing couple,
Therefore,

$$\frac{v^2}{Rr_w} (2I_w + GI_e) \cos \theta + \frac{mv^2}{R} h \cos \theta = mgh \sin \theta$$



- from the above equation, the value of angle of heel (θ) may be determined, so that the vehicle does not skid. Also, for the given value of θ , the maximum vehicle speed in the turn with out skid may be determined

Problem 2. Each wheel of a motorcycle is of 600 mm diameter and has a moment of inertia of 1.2 kg-m^2 . The total mass of the motorcycle and the rider is 180 kg and the combined centre of mass is 580 mm above the road level when motorcycle is upright. The moment of inertia of the rotating parts of the engine is 0.2 kg-m^2 . The engine speed is in the same sense. Determine the angle of heel necessary when the motorcycle takes a turn of 35 m radius at a speed of 54 km/h.

Solution: Moment of inertia of each wheel $I_w = 1.2 \text{ kg-m}^2$

Mass of the vehicle with rider is $m = 180 \text{ kg}$

An effective diameter of $d = 600 \text{ mm}$; thus radius $r = 0.3 \text{ m}$, $R = 35 \text{ m}$, $v = 54 \text{ km/h}$

Moment of inertia of engine $I_E = 0.2 \text{ kg-m}^2$

Centre of the mass is $h = 580 \text{ mm}$ above the road level = 0.58 m

Gyroscopic couple:

$$C_G = (2I_w + GI_E) \frac{v^2}{rR} \times \cos \theta = (2 \times 1.2 + 5 \times 0.2) \times \frac{(15)^2}{0.3 \times 35} \times \cos \theta = 72.86 \cos \theta$$

Centrifugal couple:

$$C_c = m \frac{v^2}{R} \times h \cos \theta = 180 \times \frac{(15)^2}{35} \times 0.58 \times \cos \theta = 671.14 \cos \theta$$

Total overturning couple: $72.86 \cos \theta + 671.14 \cos \theta = 744 \cos \theta$

Right-turning couple: $mgh \sin \theta = 180 \times 9.81 \times 0.58 \sin \theta = 1024 \sin \theta$

Therefore, $744 \cos \theta = 1024 \sin \theta$

$$\tan \theta = \frac{744}{1024} = 0.727$$

$$\text{Or } \theta = 36^\circ$$

Thank You

